

Answer Key & Teacher Notes

Session 1 – Multiplication Equations • Session 2 – Division Equations

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	First 5–8 minutes of Session 1. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 • TEACHER NOTES

Multiplication Equations

TALK PROMPTS

- Why does dividing both sides by the coefficient leave the variable by itself?
- How do you know the 7 is the coefficient and not the solution?

PARTNER TASK

Form A, problems 3 and 4. One partner writes and solves Bookshelf Rows ($7b = 133$); the other checks by multiplying. Switch roles with Egg Cartons ($15e = 270$).

WATCH FOR

- Multiplying by the coefficient instead of dividing ($5x = 65$ gives 325).
- Subtracting the coefficient ($5x = 65$ gives 60).
- Dividing the coefficient by the total ($5 \div 65$) instead of the total by the coefficient.

SESSION 2 • TEACHER NOTES

Division Equations

TALK PROMPTS

- Why does multiplying both sides by the divisor get the variable alone?
- How does the total compare with the size of one share, and why does that make sense?

PARTNER TASK

Costume Sashes. A spool of ribbon is cut evenly into 9 sashes of 16 inches. One partner writes and solves $n \div 9 = 16$; the other checks by dividing. Switch roles with Class Pizza ($s \div 5 = 12$).

WATCH FOR

- Dividing again instead of multiplying ($h \div 5 = 20$ gives 4).
- Accepting a total that is smaller than one share.
- Using the wrong number of groups (homework: Kayla and 3 friends is 4 people, not 3).

ITEM ANALYSIS

depth of knowledge (DOK) for each practice item

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 • Form A	6.EE.B.7	1	2	2	1	1	2	2	3	3	1	1
S1 • Form B	6.EE.B.7	1	2	2	1	1	2	2	3	3	1	1
S2 • Form A	6.EE.B.7	1	2	2	1	1	2	2	3	3	1	2
S2 • Form B	6.EE.B.7	1	2	2	1	1	2	2	3	3	1	1

Items before the last three are Practice; then Find the Error (DOK 3) and two Test Practice items.

GUIDED NOTES

coefficient: The number that **multiplies** the variable.

inverse operations: $\times$ and $\div$ **undo** each other.

Division Property of Equality: Divide both sides by the same number (not **zero**). They stay equal.

- Division **undoes** multiplication.
- To solve $5b = 45$, divide both sides by the **coefficient**, 5.

Example: The coefficient is **6**. Divide both sides by 6: $c = 84 \div 6 = 14$. Check: $6 \times 14 = 84$. Answer: $c = 14$ chairs in each stack

YOU TRY

1. a. $x = 9$ b. $n = 13$
2. $w = 3.2 \div 4 = 0.8$. Check: $4 \times 0.8 = 3.2$.

EXIT TICKET

1. a. $x = 8$ b. $k = 1.5$
2. $8t = 96$; $t = \$12$
3. **B** (coefficient)

PRACTICE FORM A

1. a. $k = 5$ b. $m = 13$ c. $w = 8$ d. $y = 2.5$ e. $t = 6$ f. $r = 3.5$
2. a. $5p = 175$ b. $p = 175 \div 5 = \$35$ c. $5 \times 35 = 175$
3. $7b = 133$; $b = 19$ books
4. $e = 270 \div 15 = 18$ eggs
5. $m = 154 \div 14 = 11$ miles
6. $24w = 336$; $w = \$14$
7. $18c = 216$; $c = 12$ cups. Check: $18 \times 12 = 216$.
8. a. $24r = 648$; $r = 27$ rows b. $17 \times 24 = 408$ seats c. Multiply: $24 \times 27 = 648$, so the answer checks.
9. Multiplying does not undo $\times 5$. She should divide both sides by 5: $x = 65 \div 5 = 13$. Check: $5 \times 13 = 65$.
10. **C** ($n = 12$)
11. **B** (coefficient)

Self-check word **PRODUCT**

PRACTICE FORM B

1. a. $k = 6$ b. $m = 17$ c. $w = 7$ d. $y = 3.5$ e. $t = 4$ f. $r = 4.5$
2. a. $6p = 258$ b. $p = 258 \div 6 = \$43$ c. $6 \times 43 = 258$
3. $8b = 168$; $b = 21$ books
4. $e = 264 \div 12 = 22$ eggs
5. $m = 208 \div 16 = 13$ miles
6. $28w = 420$; $w = \$15$
7. $16c = 224$; $c = 14$ cups. Check: $16 \times 14 = 224$.
8. a. $26r = 754$; $r = 29$ rows b. $19 \times 26 = 494$ seats c. Multiply: $26 \times 29 = 754$, so the answer checks.
9. Multiplying does not undo $\times 6$. He should divide both sides by 6: $x = 78 \div 6 = 13$. Check: $6 \times 13 = 78$.
10. **B** ($n = 12$)
11. **C** (divide both sides by the coefficient)

Self-check word **DIVIDES**

GUIDED NOTES

divisor: The number you **divide** by.

Multiplication Property of Equality: Multiply **both** sides by the same number. They stay equal.

equal groups: Groups that each have the **same** amount.

- Multiplication **undoes** division.
- To solve $n \div 4 = 7$, multiply both sides by **4**.

Example: Multiply both sides by **9**. $t = 7 \times 9 = \mathbf{63}$.

Check: $63 \div 9 = \mathbf{7}$. Answer: The bag holds **63** ounces.

YOU TRY

1. a. $x = 36$ b. $n = 56$
2. $n = 1.5 \times 2 = 3$. Check: $3 \div 2 = 1.5$.

EXIT TICKET

1. a. $n = 48$ b. $k = 10$
2. $r \div 5 = 3.5$; $r = 17.5$ m
3. **B** (multiply both sides by 5)

PRACTICE FORM A

1. a. $k = 63$ b. $p = 60$ c. $m = 96$ d. $n = 78$ e. $y = 9$ f. $w = 36$
2. a. $s \div 5 = 12$ b. $s = 12 \times 5 = 60$ slices c. $60 \div 5 = 12$
3. $g \div 4 = 23.50$; $g = \$94$
4. $n = 16 \times 9 = 144$ inches
5. $s = 14 \times 15 = 210$ seeds
6. $w \div 12 = 16$; $w = 192$ liters
7. $p \div 7 = 24$; $p = 168$ pages
8. a. $t \div 14 = 45$; $t = 630$ tiles b. $18 \times 45 = 810$ tiles c. Divide: $630 \div 14 = 45$, so it checks.
9. Dividing again does not undo $\div 5$. He should multiply both sides by 5: $h = 20 \times 5 = 100$. Check: $100 \div 5 = 20$.
10. **A** ($x = 55$)
11. **C** (greater than 20)

Self-check word DIVISOR

HOMEWORK PACKET • ANSWERS

same answers for the core packet and Level 1

SHEET 1 • MULTIPLICATION EQUATIONS

1. a. $x = 9$ b. $n = 8$ c. $y = 12$ d. $k = 7$ e. $m = 17$ f. $w = 1.5$
2. $5c = 65$; $c = 13$
3. $5f = 11.25$; $f = \$2.25$. Check: $5 \times 2.25 = 11.25$.
4. $6s = 84$; $s = 14$ seedlings. Check: $6 \times 14 = 84$.
5. $4n = 80$ has the greater solution. The same total split into fewer equal groups makes each group larger. Check: $4n = 80$ gives $n = 20$; $5n = 80$ gives $n = 16$.

PRACTICE FORM B

1. a. $k = 72$ b. $p = 70$ c. $m = 66$ d. $n = 84$ e. $y = 10$ f. $w = 48$
2. a. $s \div 6 = 11$ b. $s = 11 \times 6 = 66$ slices c. $66 \div 6 = 11$
3. $g \div 3 = 31.50$; $g = \$94.50$
4. $n = 18 \times 7 = 126$ inches
5. $s = 15 \times 13 = 195$ seeds
6. $w \div 14 = 15$; $w = 210$ liters
7. $p \div 6 = 26$; $p = 156$ pages
8. a. $t \div 16 = 35$; $t = 560$ tiles b. $20 \times 35 = 700$ tiles c. Divide: $560 \div 16 = 35$, so it checks.
9. Dividing again does not undo $\div 6$. She should multiply both sides by 6: $m = 12 \times 6 = 72$. Check: $72 \div 6 = 12$.
10. **C** ($n = 72$)
11. **B** (multiply both sides by the divisor)

Self-check word PRODUCT

SHEET 2 • DIVISION EQUATIONS

1. a. $n = 42$ b. $y = 65$ c. $k = 45$
2. $t \div 3 = 17$; $t = 51$
3. $r \div 8 = 2.5$; $r = 20$ meters
4. $z \div 6 = 2$; $z = 12$ slices. Check: $12 \div 6 = 2$.
5. The cards are shared by 4 people (Kayla and 3 friends), not 3. Correct equation: $c \div 4 = 9$, so $c = 36$ cards.
6. **D** ($y = 54$)
7. **A** ($p \div 12 = 5$)